

B.Tech. Degree IV Semester Regular/Supplementary Examination June 2024

CS 19-202-0405 DATA STRUCTURES AND ALGORITHMS

(2019 Scheme)

Time: 3 Hours

Maximum Marks: 60

Course Outcome

On successful completion of the course, the students will be able to:

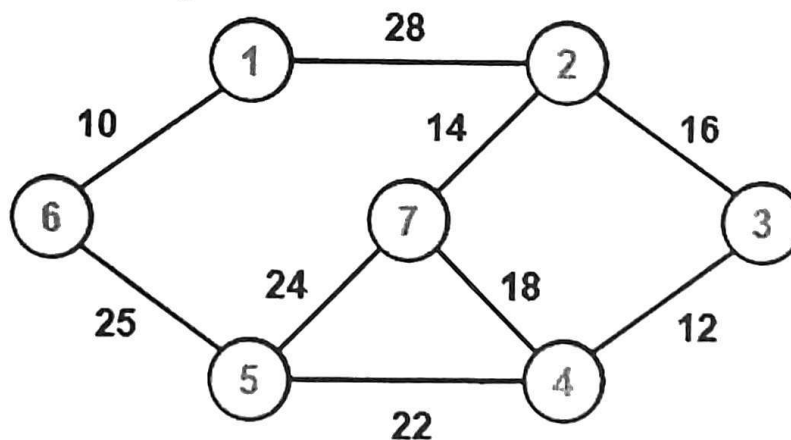
- CO1: Explain the important features of data structures like arrays, linked lists, trees and graphs.
 CO2: Define advanced data structures such as balanced search trees, hash tables, spatial data structures etc.
 CO3: Create the different data structures to solve a problem.
 CO4: Describe and compare the performance of various sorting algorithms like quicksort, mergesort and heapsort.
 CO5: Describe algorithms on trees and graphs such as traversals, shortest path and minimum spanning tree.
 CO6: Design a data structure and algorithm for maximum efficiency.

Bloom's Taxonomy Levels (BL): L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 – Create

PO – Programme Outcome

PART A (Answer *ALL* questions)

	(8 × 3 = 24)	Marks	BL	CO	PO
I. (a) ✓ Explain about linear and nonlinear data structures.	3	L1	1	1	
(b) ✓ Illustrate bubble sort algorithm on the following input [30, 20, 10, 60, 70, 40]	3	L2	1	1	
(c) Explain polish and reverse polish notation? Give examples for each.	3	L1	1	1	
(d) ✓ Explain about the representation of header node in linked list.	3	L2	1	1	
(e) ✓ Consider a binary tree T that has 180 leaf nodes. Find the number of nodes in T that has exactly two children.	3	L1	2	1	
(f) ✓ Illustrate how you represent a Binary Tree in memory using array.	3	L1	2	1	
(g) ✓ Give any two representations of graph.	3	L2	3	3	
(h) ✓ Construct the minimum spanning tree (MST) for the given graph using Prim's algorithm.	3	L2	4	2,3	



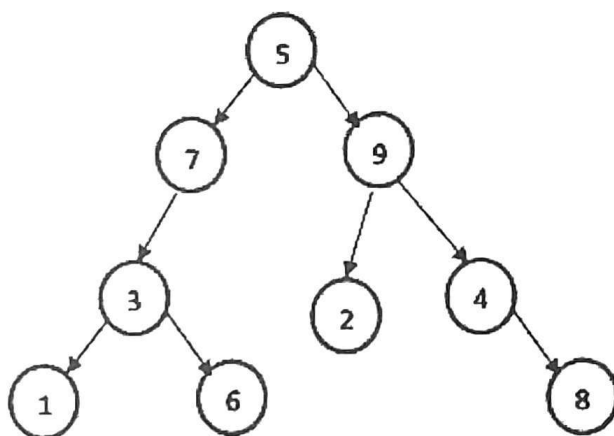
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PART B

(4 × 12 = 48)

		Marks	BL	CO	PO
		12	L1	1	1
II.	Write an algorithm/pseudocode to sort elements using heap sort technique. Illustrate the working of heap sort algorithm on the following input 15, 20, 7, 9, 30.				
	OR				
III.	(a) Define hashing. What are the properties of a good hash function?	4	L1	1	1
	(b) Define collision. What is linear probing? The following keys 43, 165, 62, 123, 142 are inserted into an initially empty hash table of length 10 using hash function $h(k) = k \bmod 10$ and linear probing. What is the resultant hash table?	8	L1	2	1
IV.	(a) Compare and contrast between singly linked list and doubly linked lists.	4	L1	2	1
	(b) Write code sequences for inserting elements in a linked list at beginning, last and at a particular position.	8	L1	2	1
	OR				
V.	(a) Assume that a stack is represented using linked list. Write algorithms for the following operations (i) Push (). (ii) Pop ().	8	L1	2	1
	(b) Write the Algorithms to evaluate postfix expression. Trace the algorithm on the following input. 10 5 + 60 6 / * 8-	4	L1	2	1
VI.	(a) Explain tree traversals? Write algorithm for each.	7	L3	3	3
	(b) What is the output obtained after preorder, inorder and postorder traversal of the following tree.	5	L3	3	3

**OR**

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|------|-----|---|---|----|---|---|
| VII. | (a) | What is a Binary Search Tree (BST)? Show the structure of the binary search tree after adding each of the following values in that order. What is the height of the created binary search tree?
50, 15, 62, 5, 20, 58, 91, 3, 8, 37, 60, 24. | 6 | L1 | 3 | 1 |
| | (b) | Write an algorithm/pseudocode to perform the following operations on a binary search tree.
(i) Delete the node containing 37
(ii) Delete the node containing 62 from BST. Write pseudocode for both operations. | 6 | L2 | 3 | 1 |

(Continued)

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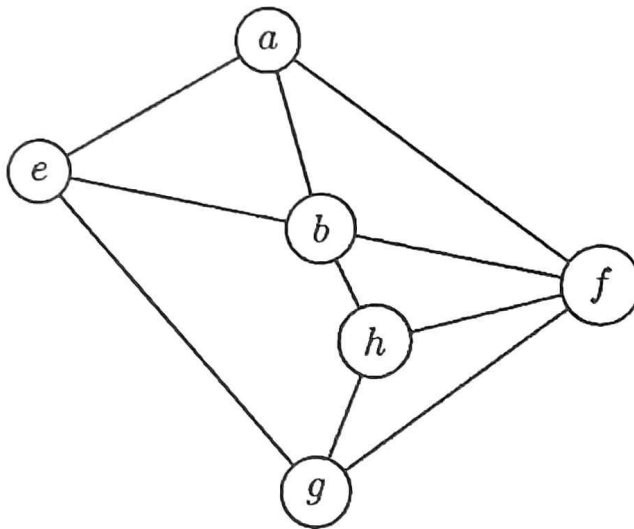
Marks	BL	CO	PO
8	L2	4	1

VIII. (a) ✓ Write the algorithm for DFS traversal. Consider the following graph among the following sequences:

- (i) abeghf
- (ii) abfehg
- (iii) abfhge
- (iv) afghbe.

Which are the depth-first traversals of the above graph? Justify your answer.

- (i) I, II and IV only
- (ii) I and IV only
- (iii) II, III and IV only
- (iv) I, III and IV only



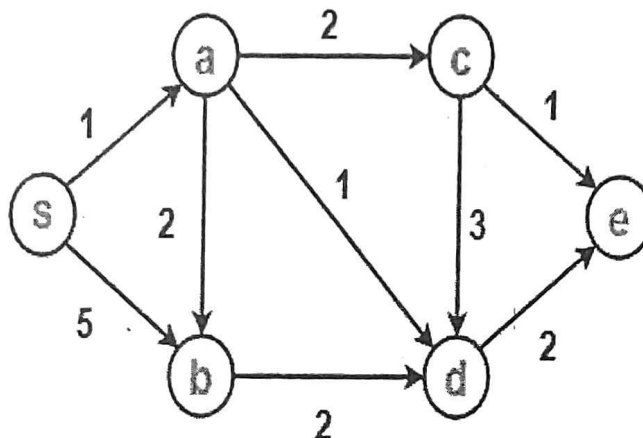
(b) ✓ Differentiate between Kruskal's and Prim's algorithms for the construction of minimum spanning tree.

4	L2	4	1
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OR

IX. (a) Write the Dijkstra's Algorithm, using Dijkstra's Algorithm, find the shortest distance from source vertex 'S' to remaining vertices in the following graph.

8	L2	4	1
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(b) What is the drawback of B - trees? How it is compensated in B+ - trees?

4	L2	4	1
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Blooms's Taxonomy Level's

L1 – 42%, L2 – 31.5%, L3 – 26.5%.
